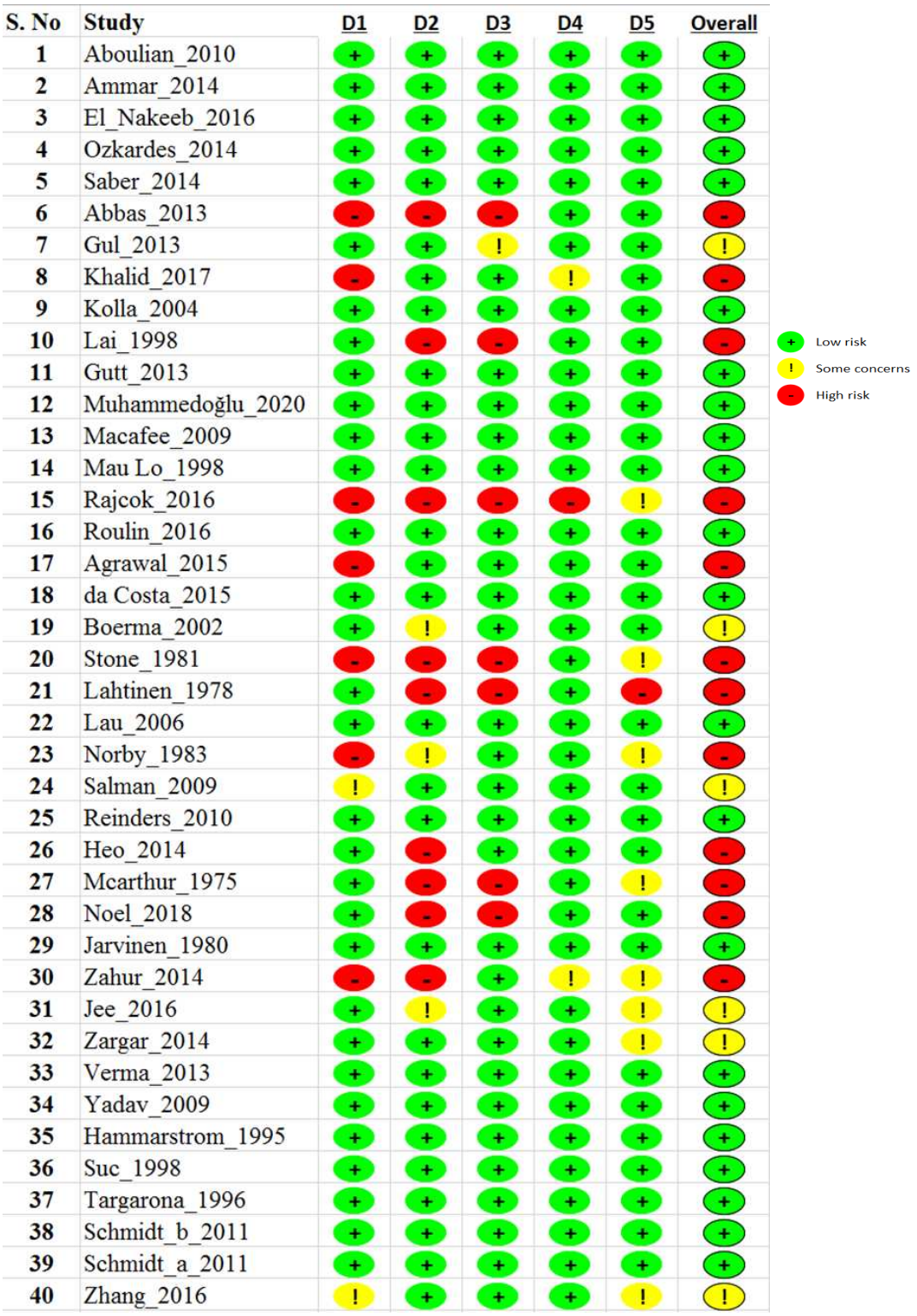
Online Supplementary Material

- Supplementary Figures – **Page 2-14**
- Supplementary Tables - **Page 15-24**
- Prisma checklist - **Page 25-26**

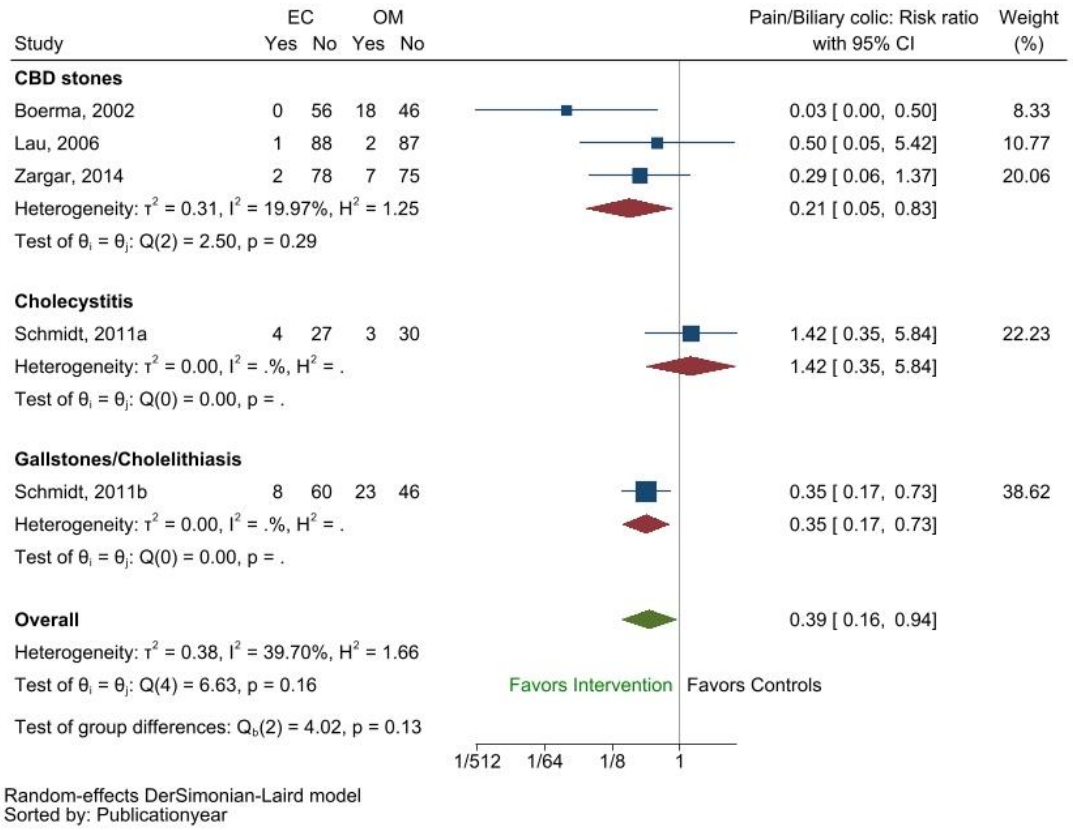
Bagepally BS, *et al. BMJ Open Gastro* 2021; 8:e000675. doi: 10.1136/bmjgast-2021-000675

Supplementary Figures



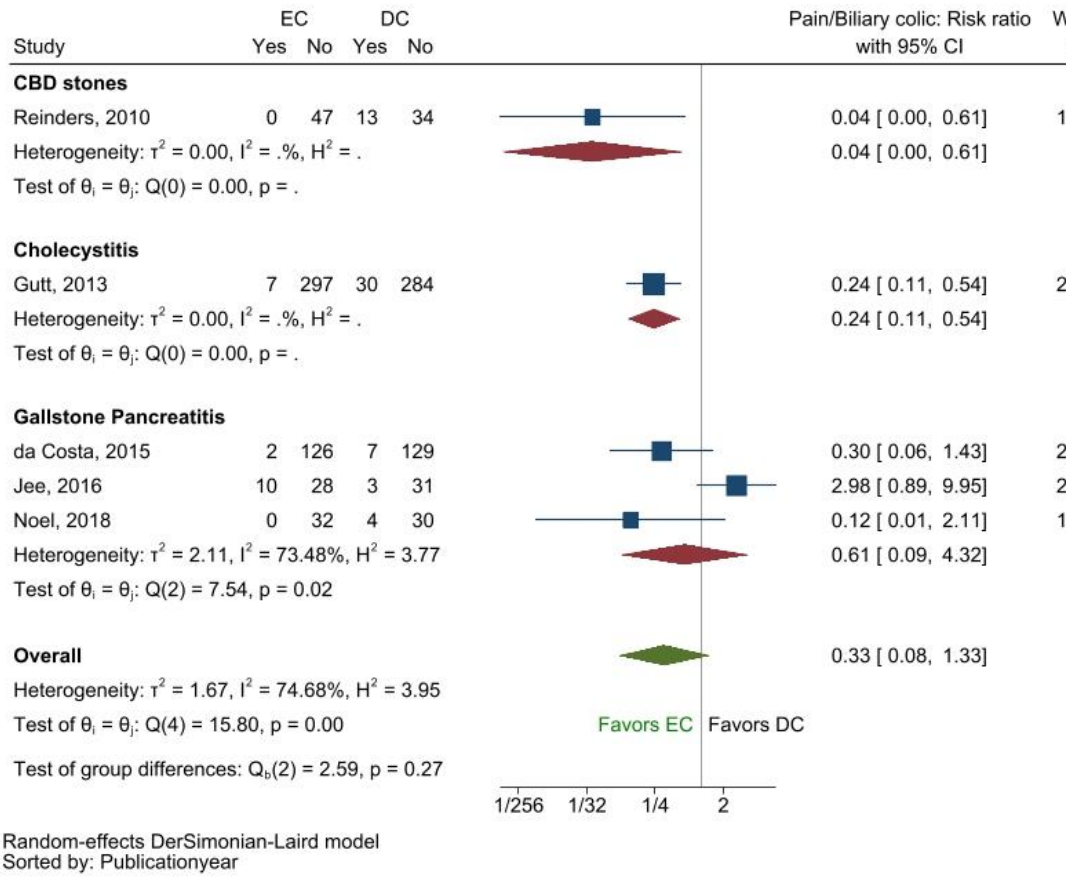
Supplementary Figure 1: Assessment of Risk of Bias
Bias due to D1-Randomization process; D2-Deviation from intended intervention;
D3-Missing outcome data; D4- Measurement of outcome; D5-Selection of reported result.

Supplementary Figures



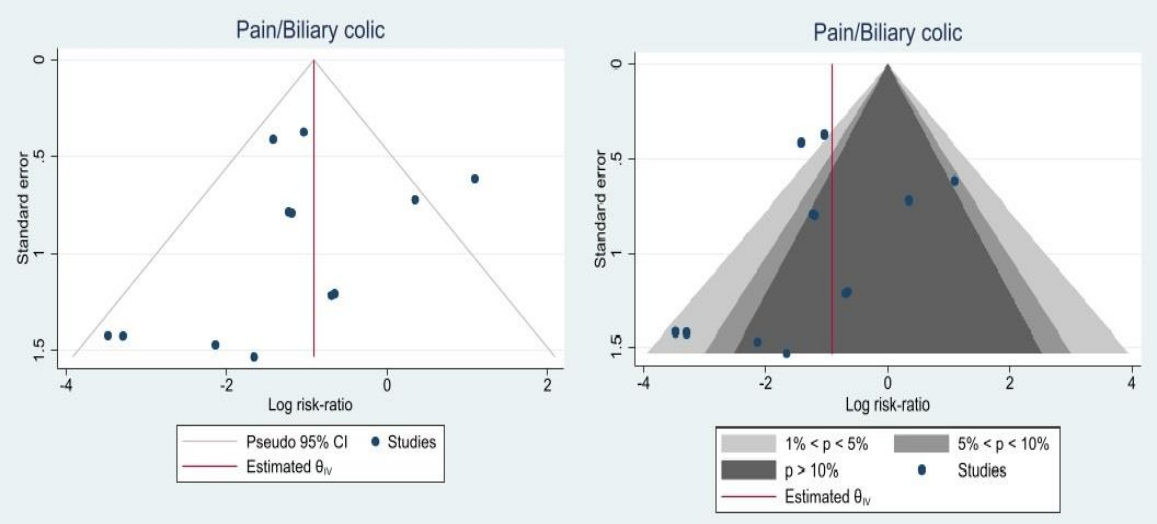
Supplementary Figure 2: Subgroup analysis of Pain/Biliary Colic in early cholecystectomy (EC) versus observation management (OM) by Baseline gallstone complication.

Supplementary Figures



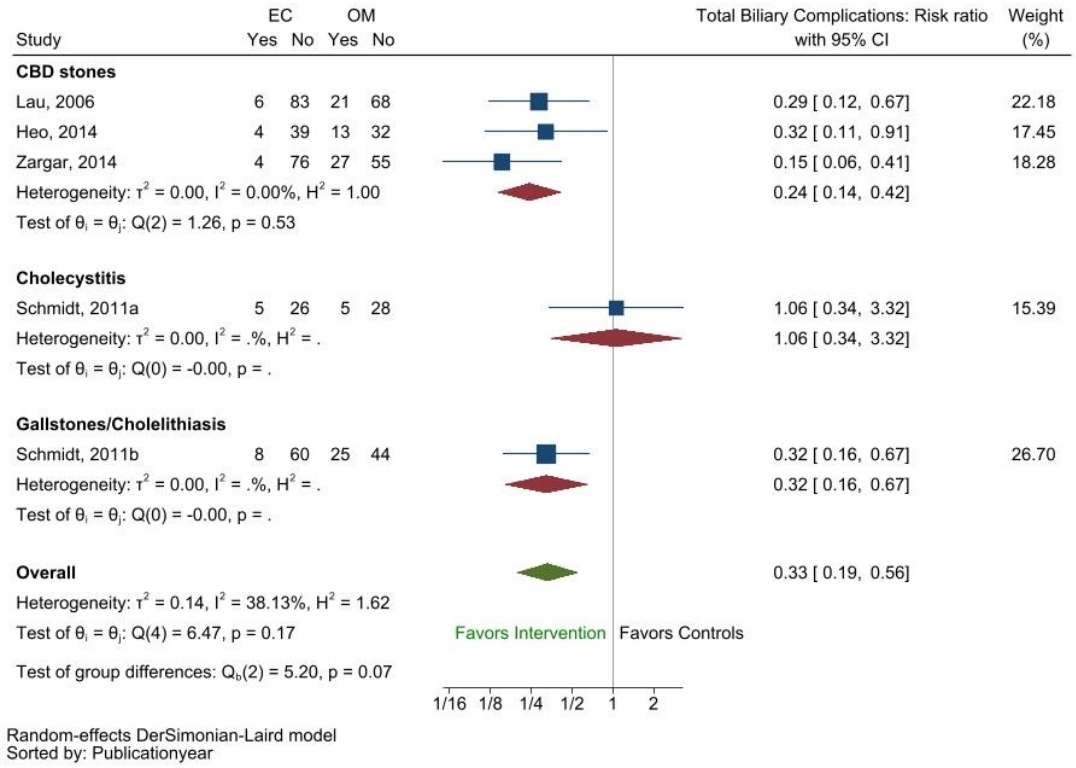
Supplementary Figure 3: Subgroup analysis of Pain/Biliary Colic in Early Cholecystectomy (EC) versus Delayed Cholecystectomy (DC) by Baseline gallstone complication.

Supplementary Figures



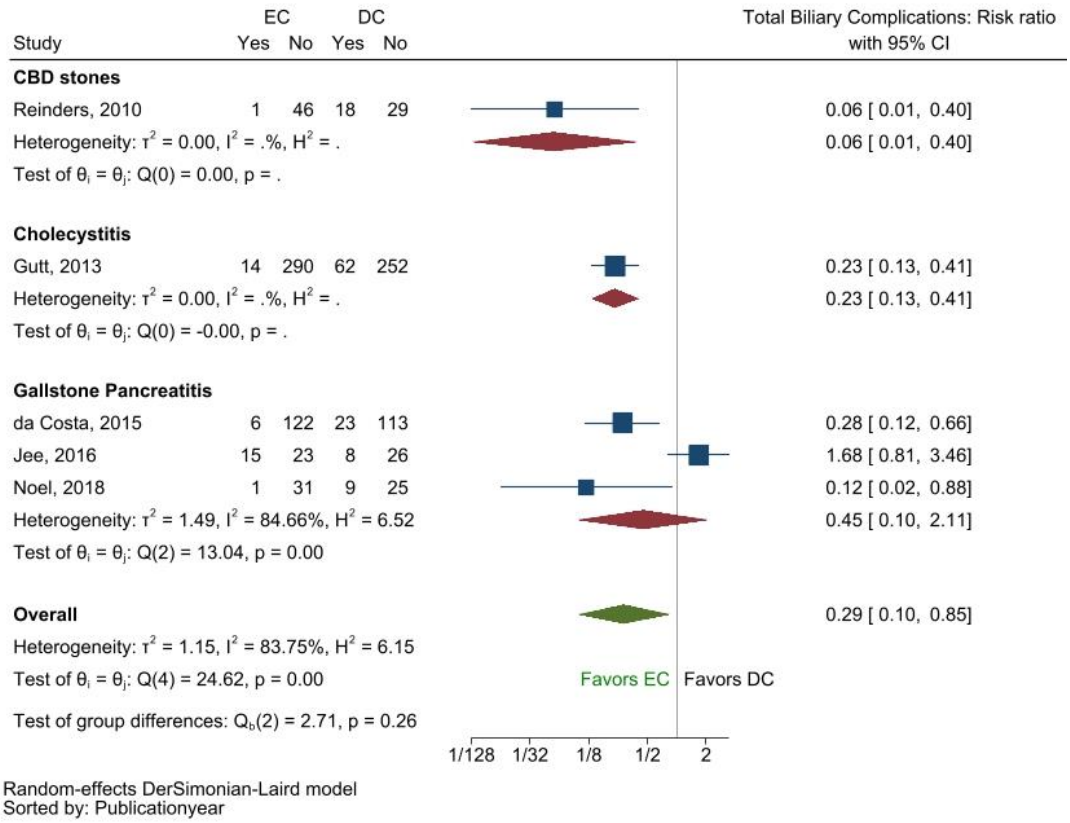
Supplementary Figure 4: Funnel and contour-enhanced funnel plots of early versus delayed cholecystectomy for Pain/Biliary Colic

Supplementary Figures



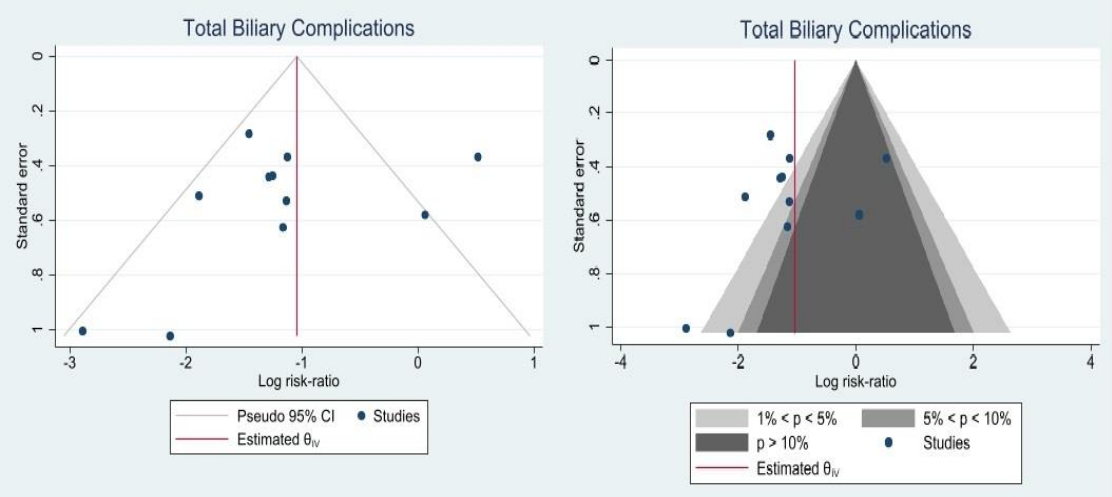
Supplementary Figure 5: Subgroup analysis of Total biliary complications in early cholecystectomy (EC) versus observation management (OM) by Baseline gallstone complication.

Supplementary Figures



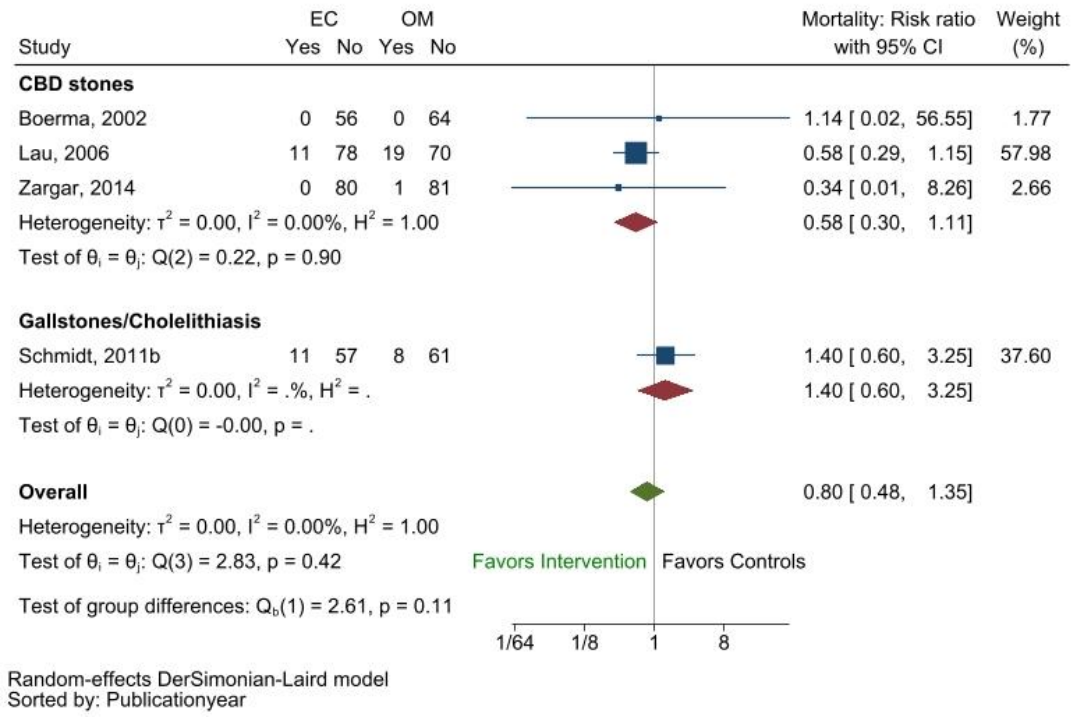
Supplementary Figure 6: Subgroup analysis of Total biliary complications in Early Cholecystectomy (EC) versus Delayed Cholecystectomy (DC) by Baseline gallstone complication.

Supplementary Figures



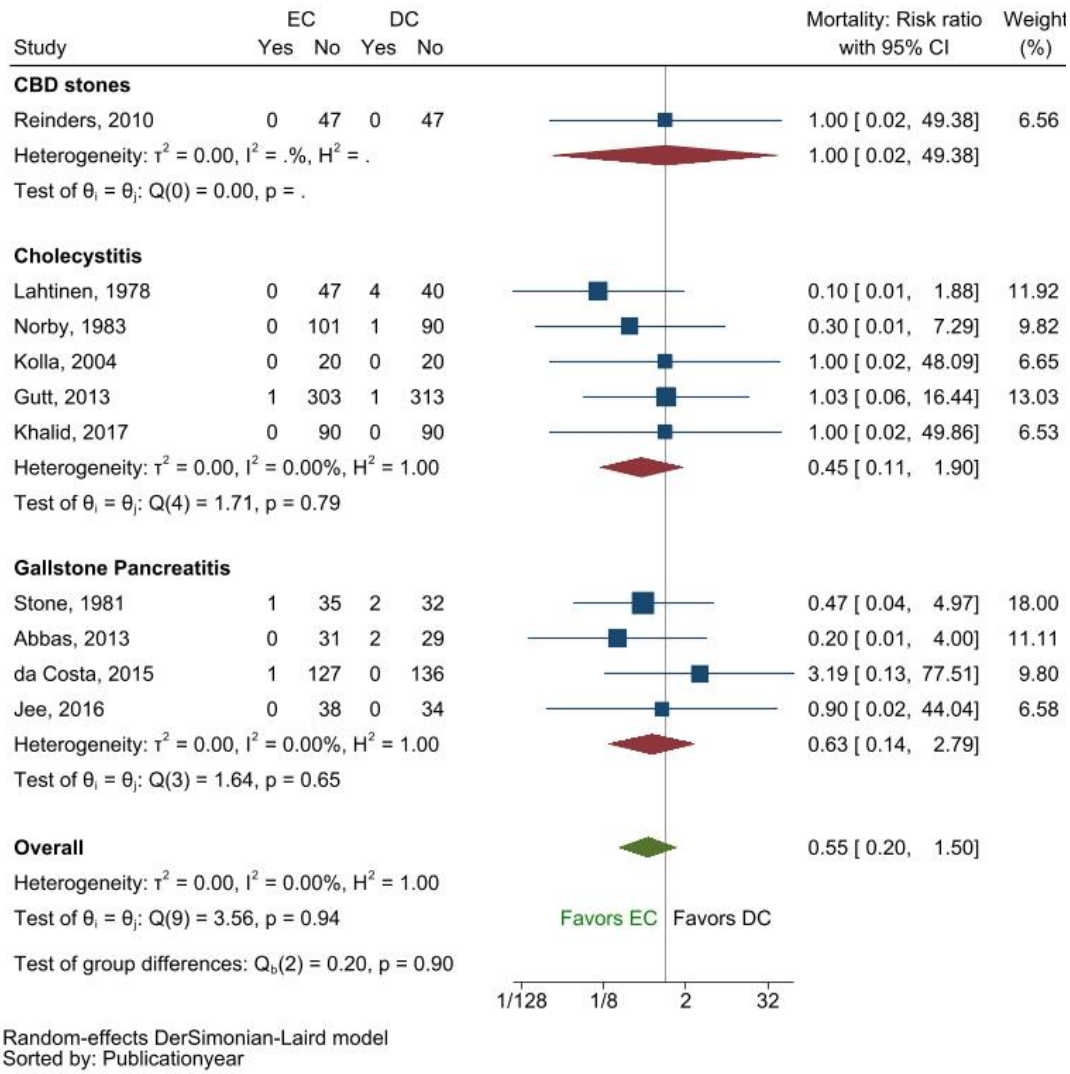
Supplementary Figure 7: Funnel and contour-enhanced funnel plots of early cholecystectomy versus entirely/relatively conservative management for Total Biliary Complications

Supplementary Figures



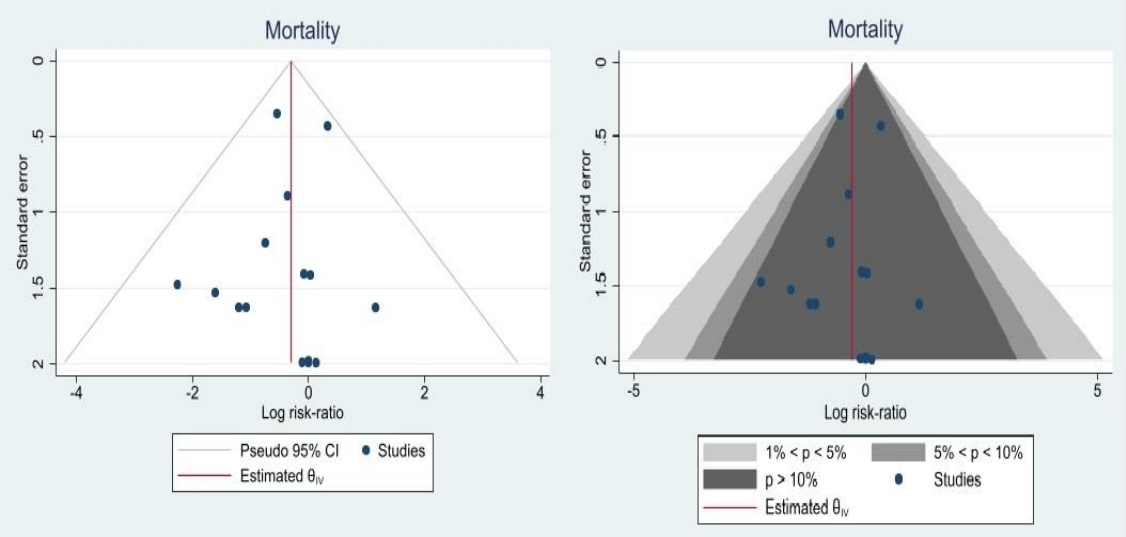
Supplementary Figure 8: Subgroup analysis of Mortality in early cholecystectomy (EC) versus observation management (OM) by Baseline gallstone complication.

Supplementary Figures



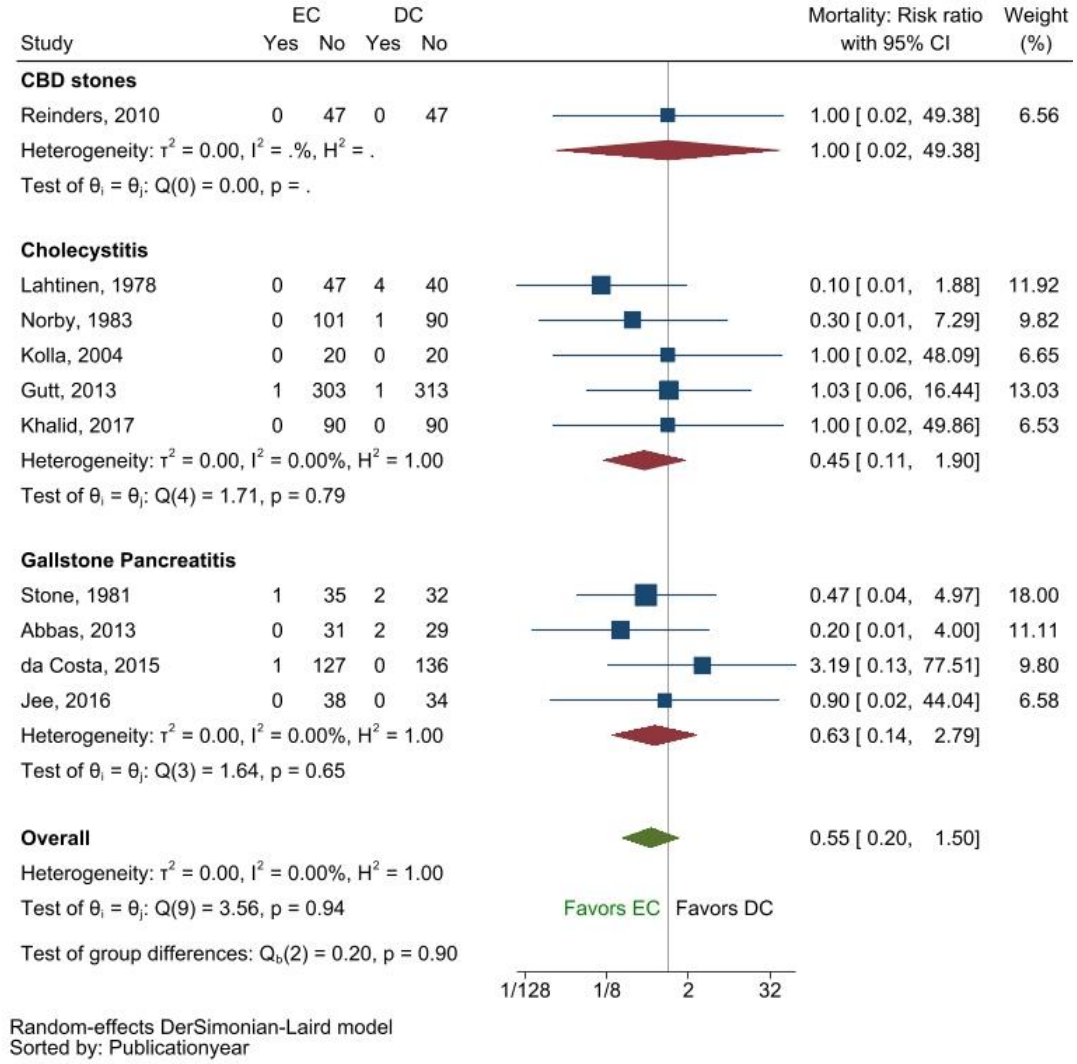
Supplementary Figure 9: Subgroup analysis of Mortality in Early Cholecystectomy (EC) versus Delayed Cholecystectomy (DC) by Baseline gallstone complication.

Supplementary Figures



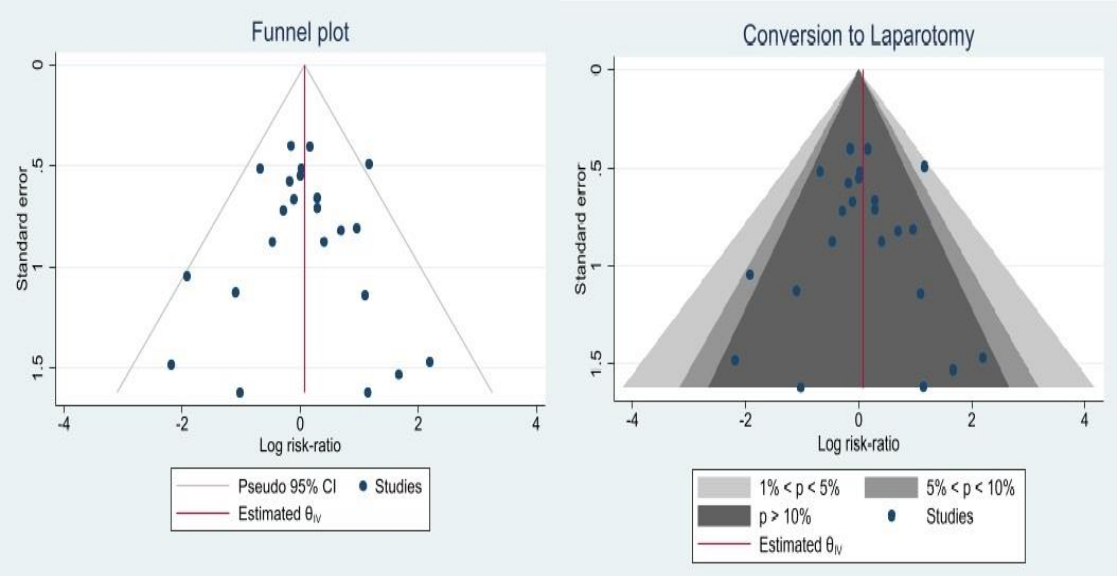
Supplementary Figure 10: Funnel and contour-enhanced funnel plots of early cholecystectomy versus conservative management for Mortality

Supplementary Figures



Supplementary Figure 11: Subgroup analysis of Conversion to laparotomy in Early Cholecystectomy (EC) versus Delayed Cholecystectomy (DC) by Baseline gallstone complication.

Supplementary Figures



Supplementary Figure 12: Funnel and contour-enhanced funnel plots of early versus delayed cholecystectomy for Conversion to Laparotomy

Supplementary Tables

Supplementary Table 1 Characteristics of included studies

Study	Country	Study Design	Baseline gallstone complication	Type of surgery	Groups (Sample size)	Age (Mean)	Gender (%)	Duration of symptoms (hours)	Timing of surgery
Muhammedoğlu et al 2020	Turkey	RCT-SC	CBD stones	LC	EC (n=82) DC (n=37)	61.5 53.5	- -	- -	Not Specified 6-8 weeks
Noel et al 2018	Sweden	RCT-SC	Gallstone Pancreatitis	LC	EC (n=32) DC (n=34)	48 43.5	62.5% 55.9%	- -	within 48 hours 6 weeks
Khalid et al 2017	Pakistan	RCT-SC	Cholecystitis	LC	EC (n=90) DC (n=90)	- -	- -	- -	within 72 hours 6-8 weeks
El Nakeeb et al 2016	Egypt	RCT-SC	CBD stones	LC	EC (n=55) DC (n=55)	43 47	70.9% 65.5%	- -	within 72 hours after 1 month
Rajcok et al 2016	Slovakia	RCT-SC	Cholecystitis	LC	EC (n=31) DC (n=31)	56.3 59.8	48.30% 45.10%	- -	within 72 hours 6-8 weeks
Roulin et al 2016	Switzerland	RCT-SC	Cholecystitis	LC	EC (n=42) DC (n=44)	55.8 57.9	43% 43%	96 96	within 72 hours after 6 weeks
Zhang et al 2016	China	RCT-SC	Gallstone Pancreatitis	LC	EC (n=49) DC (n=53)	62.1 63.5	57.1% 62.26%	- -	within 7 days >7 days
Jee et al 2016	Malaysia	RCT-SC	Gallstone Pancreatitis	LC	EC (n=38) DC (n=34)	42.5 42.5	52.63% 61.76%	- -	within 7 days 6-8 weeks
da Costa et al 2015	Netherlands	RCT-MC	Gallstone Pancreatitis	LC	EC (n=128) DC (n=136)	53 54	59% 62%	- -	within 72 hours 25-30 days
Agrawal et al 2015	India	RCT-SC	Cholecystitis	LC	EC (n=25) DC (n=25)	47.28 50.96	68% 68%	35.44 36.8	within 24 hours 6-8 weeks
Ammar et al 2014	Egypt	RCT-SC	CBD stones	LC	EC (n=31) DC (n=29)	46.2 47.3	67.70% 724%	- -	within 24 hours >24 hours
3Heo et al 2014	Korea	RCT-SC	CBD stones	LC	CH (n=43) CM (n=45)	64.02 63.96	41.9% 44.4%	- -	- -
Zahur et al 2014	Pakistan	RCT-SC	Cholecystitis	LC	EC (n=47) DC (n=41)	- -	- -	- -	within 48 hours 6-8 weeks

Supplementary Tables

Study	Country	Study Design	Baseline gallstone complication	Type of surgery	Groups (Sample size)	Age (Mean)	Gender (%)	Duration of symptoms (hours)	Timing of surgery
Zargar et al 2014	India	RCT-SC	CBD stones	-	CH (n=80)	78.2	-	-	-
					CM (n=82)	77.3	-	-	-
Gul et al 2013	India	RCT-SC	Cholecystitis	LC	EC (n=30)	39.83	-	-	within 72 hours
					DC (n=30)	38.27	-	-	6-12 weeks
Verma et al 2013	India	RCT-SC	Cholecystitis	LC	EC (n=30)	31.73	86.6%	-	within 72 hours
					DC (n=30)	32.8	93.3%	-	6-8 weeks
Schmidt et al a et al 2011	Norway	RCT-MC	Cholecystitis	LC	CH (n=31)	55.5	60.6%	-	-
					CM (n=33)	61	54.8%	-	-
Schmidt et al b et al 2011	Norway	RCT-MC	Uncomplicated gallstones	LC	CH (n=68)	52	80.9%	-	-
					CM (n=69)	54	82.6%	-	-
Aboulian et al 2010	USA	RCT-SC	Gallstone Pancreatitis	LC	EC (n=25)	33	8%	48	within 48 hours
					DC (n=25)	41	12%	48	Not specified
Reinders et al 2010	Netherlands	RCT-MC	CBD stones	LC	EC (n=47)	55	76.5%	-	within 72 hours
					DC (n=47)	47	61.7%	-	6-8 weeks
Salman et al 2009	Turkey	RCT-SC	CBD stones	LC	EC (n=39)	43.5	66.60%	-	within 48 hours
					DC (n=40)	44.6	70%	-	within 72hours to 7 days
Yadav et al 2009	Nepal	RCT-SC	Cholecystitis	LC	EC (n=25)	42.68	-	-	within 48 hours
					DC (n=25)	40.26	-	-	6-8 weeks
Kolla et al 2004	India	RCT-SC	Cholecystitis	LC	EC (n=20)	41.5	85%	35.1	within 24 hours
					DC (n=20)	38.6	75%	36.1	6-12 weeks
Lai et al 1998	Hong Kong	RCT-SC	Cholecystitis	LC	EC (n=53)	55.8	56.60%	-	within 24 hours
					DC (n=51)	56.1	70.50%	-	6-8 weeks
Mau Lo et al 1998	Hong Kong, China	RCT-SC	Cholecystitis	LC	EC (n=45)	59	42.20%	48	within 72 hours
					DC (n=41)	61	48.70%	48	8-12 weeks
Norby et al 1983	Swedon	RCT-MC	Cholecystitis	OC	EC (n=101)	58	65.30%	-	within 7 days
					DC (n=91)	58	61.50%	-	Not specified
Jarvinen et al 1980	Finland	RCT-SC	Cholecystitis	OC	EC (n=80)	57.8	50%	52.8	within 7 days
					DC (n=75)	56.7	52%	55.2	8-16 weeks

Supplementary Tables

Study	Country	Study Design	Baseline gallstone complication	Type of surgery	Groups (Sample size)	Age (Mean)	Gender (%)	Duration of symptoms (hours)	Timing of surgery
Mcarthur et al 1975	United Kingdom	RCT-SC	Cholecystitis	OC	EC (n=15) DC (n=17)	48.9 49.8	82.3% 93.3%	38.1 47.9	Not Specified 8-12 weeks
Ozkardes et al 2014	Turkey	RCT-SC	Cholecystitis	LC	EC (n=30) DC (n=30)	58.03 59.43	66.70% 56.70%	- -	within 24 hours 6-8 weeks
Saber et al 2014	Egypt	RCT-MC	Cholecystitis	LC	EC (n=60) DC (n=60)		75% 70%	- -	within 72 hours 6-8 weeks
Abbas et al 2013	Pakistan	RCT-SC	Gallstone Pancreatitis	OC	EC (n=31) DC (n=31)	40.22 53.51	83.80% 83.80%	- -	Not Specified 6-8 weeks
Gutt et al 2013	Germany	RCT-MC	Cholecystitis	LC	EC (n=304) DC (n=314)	55.6 56.8	62.8% 54.8%	- -	within 24 hours 7-45 days
Macafee et al 2009	United Kingdom	RCT-SC	Cholecystitis	LC	EC (n=36) DC (n=36)	52 53	72.20% 58.30%	- -	within 72 hours after 3 months
Lau et al 2006	China	RCT-SC	CBD stones	LC	CH (n=89) CM (n=89)	43 49	51.7% 44.9%	- -	- -
Boerma et al 2002	Netherlands	RCT-MC	CBD stones	LC	CH (n=56) CM (n=64)	60 63	59% 51%	- -	- -
Suc et al 1998	France	RCT-MC	CBD stones	OC	CH (n=105) EM (n=97)	66.7 66.8	68.50% 68.04%	- -	- -
Targarona et al 1996	Spain	RCT-SC	CBD stones	OC	CH (n=48) EM (n=50)	80 79	68.75% 70%	- -	- -
Hammarstrom et al 1995	Sweden	RCT-SC	CBD stones	OC	CH (n=41) EM (n=39)	73.5 75	- -	- -	- -
Stone et al 1981	Georgia	RCT-SC	Gallstone Pancreatitis	OC	EC (n=36) DC (n=34)	- -	- -	- -	within 72 hours 3-6 months
Lahtinen et al 1978	Finland	RCT-SC	Cholecystitis	OC	EC (n=47) DC (n=44)	63.8 62.8	60% 60%	- -	within 7 days 8-12 weeks

RCT-Randomized Control Trial; MC-Multicentre; SC-Single Centre; LC-Laparoscopic Cholecystectomy; OC- Open Cholecystectomy; EC- Early Cholecystectomy; DC- Delayed Cholecystectomy; CM-Conservative Management; EM- Endoscopic Management; Ch-Cholecystectomy; CBD- Common Bile Duc

Supplementary Tables

Supplementary Table 2: Subgroup analysis of surgical complications of early cholecystectomy (EC) versus delayed cholecystectomy (DC) based on timing of surgery

Surgical Complications	Timing of surgery	RR (95% CI)	I^2 (%)	Q	References
Total Intra-operative complications	EC within 7 days (2)	1.21 (0.15 to 9.65)	18.99%	1.23	13 24
	EC within 72 hours (3)	0.95 (0.28 to 3.20)	61.33%	5.17	11 25 26
	DC <6 weeks (1)	1.02 (0.02 to 50.42)	-	-	25
	DC >6 weeks (3)	0.83 (0.24 to 2.84)	64.54%	5.64	11 24 26
	Overall (5)	1.00 (0.40 to 2.51)	38.64%	6.52	
Total Post-operative complications	EC within 7 days (1)	0.89 (0.19 to 4.14)	-	-	24
	EC within 72 hours (11)	0.94 (0.68 to 1.29)	0%	8.01	1 5 8 9 18-22 25 26
	DC <6 weeks (1)	0.26 (0.03 to 2.19)	-	-	25
	DC >6 weeks (11)	0.97 (0.70 to 1.32)	0%	6.58	1 5 8 9 18-22 25 26
	Overall (12)	0.94 (0.69 to 1.28)	0%	8.02	
Total Surgical complications	EC within 7 days (3)	0.83 (0.53 to 1.31)	0%	0.16	7 13 24
	EC within 72 hours (12)	1.35 (1.02 to 1.79)	0%	9.52	1 6 8 9 14 17 19-21 23 26 28
	DC <6 weeks (2)	0.98 (0.50 to 1.91)	0%	0.09	6 14
	DC >6 weeks (14)	1.23 (0.94 to 1.62)	3.51%	13.47	1 7-9 15 17 19-21 23 24 26-28
	Overall (17)	1.15 (0.91 to 1.45)	0%	14.63	
Mortality	EC within 7 days (3)	0.25 (0.04 to 1.62)	0%	0.78	12 13 24
	EC within 72 hours (6)	0.97 (0.27 to 3.51)	0%	0.9	6 8 9 14 21 23
	DC <6 weeks (2)	1.67 (0.21 to 13.55)	0%	0.27	6 14
	DC >6 weeks (7)	0.59 (0.21 to 1.69)	0%	3.40	8 9 12 21 23 24 30
	Overall (9)	0.63 (0.22 to 1.8)	0%	3.06	
Conversion to Laparotomy	EC within 7 days (2)	0.78 (0.23 to 2.63)	0%	0.27	10 24
	EC within 72 hours (18)	1.01 (0.75 to 1.37)	0%	12.97	1-5 8 9 11 14 16-18 20 22 25 26 28 29
	DC <6 weeks (5)	0.71 (0.36 to 1.43)	0%	3.46	2 3 10 14 25
	DC >6 weeks (15)	1.07 (0.78 to 1.48)	0%	8.87	1 4 5 8 9 11 16-18 20 22 24 26 28 29
	Overall (20)	1.00 (0.74 to 1.34)	0%	13.40	

Supplementary Tables

Supplementary Table 4: Risk ratios of biliary complications in early cholecystectomy versus conservative management (Observation and Endoscopic management)

Complications (no. of studies)	RR (CI)	I^2	Q	Reference
Pain/Biliary Colic (7)	0.39 (0.21 to 0.75)	13.20%	6.91	31 32 33-37
Biliary Pancreatitis (5)	0.99 (0.25 to 3.90)	0%	0.98	31 38 35-37 38
Acute cholecystitis (9)	0.09 (0.03 to 0.23)	0%	1.13	31 32 38 33-37 39
CBD stones (2)	0.35 (0.06 to 2.15)	0%	0.00	35 36
Cholangitis (6)	0.52 (0.28 to 0.97)	0%	3.76	32-34 37 38
Total Biliary complications (6)	0.33 (0.21 to 0.51)	22.7%	6.47	32 34-37 39
Mortality (6)	0.80 (0.49 to 1.30)	0%	2.86	32-34 36-38

Supplementary Tables

Supplementary Table 5. Search terms and no. of hits in PubMed

PICOS	Pubmed Search terms	**Hits (on date 16th July 2020)**
P	"gall bladder" OR gallbladder OR cholecystitis OR cholelithiasis OR gallstone OR “gall stone” OR “gall stones” OR cholecystolithiasis OR "bile duct stone" OR “common bile duct” OR CBD OR "biliary disease" OR "biliary tract disease" OR cholec* OR “gallbladder inflammation”	**151,737**
I	cholecystectomy OR "gallbladder removal" OR "gallbladder excision" OR ("surgical removal" AND ("gall bladder" OR gallbladder)) OR ("surgical excision" AND ("gall bladder" OR gallbladder)) OR "restrictive strategy" OR cholecystostomy OR "biliary tract surgery" OR "bile duct operation" OR "bile duct surgery" OR "bile tract surgery" OR "biliary surgery" OR "biliary tract operation" OR "biliary tract reoperation" OR "biliary tract surgery" OR "biliary tract surgical procedures" OR "gall bladder surgery" OR "gallbladder operation" OR "gallbladder surgery" OR sphincterotomy	**55,069**
C	“conservative treatment” OR “conservative management” OR “conservative therapy” OR “nonoperative treatment” OR “nonsurgical treatment” OR 'organ sparing treatment' OR ‘'organ sparing treatments' OR supportive OR non-invasive OR non-surgical OR non-operative OR "wait and watch" OR "wait and see" OR watchful-waiting OR "usual care"	**10,190,039**
S	"randomized controlled trial" [pt] OR "controlled clinical trial" [pt] OR randomized [tiab] OR placebo [tiab] OR "drug therapy" [sh] OR randomly [tiab] OR trial [tiab] OR groups [tiab]	**4,809,100**
PICS	**PICS**	**1742**
Additional search with new search terms	(Lithotripsy OR dissolution OR "ursodeoxycholic acid" OR "endoscopic retrograde cholangiopancreatography" OR "Percutaneous cholecystolithotomy" OR "methyl tert-butyl ether" OR ursodiol)) NOT PICS	130
Updated search	Updated search as on 12th January 2021	74
	Total	**1945**

Supplementary Tables

Supplementary Table 6. Search terms and no. of hits in Embase

PICOS	EMBASE Search terms	Hits (on date 16th July 2020)
P	'gall bladder'/exp OR 'gall bladder' OR 'gallbladder'/exp OR gallbladder OR 'cholecystitis'/exp OR cholecystitis OR 'cholelithiasis'/exp OR cholelithiasis OR 'gallstone'/exp OR gallstone OR 'gall'/exp OR gall OR 'cholecystolithiasis'/exp OR cholecystolithiasis OR 'calculi'/exp OR calculi OR 'choledocholithiasis'/exp OR choledocholithiasis OR 'bile duct stone'/exp OR 'bile duct stone' OR 'cbd stone' OR 'biliary disease'/exp OR 'biliary disease' OR 'biliary tract disease'/exp OR 'biliary tract disease' OR "gallbladder inflammation"	**356,517**
I	cholecystectomy OR 'gallbladder removal' OR 'gallbladder excision' OR ('surgical removal' AND ('gall bladder' OR gallbladder)) OR ('surgical excision' AND ('gall bladder' OR gallbladder)) OR 'restrictive strategy' OR cholecystostomy OR 'biliary tract surgery'/exp OR 'bile duct operation' OR 'bile duct surgery' OR 'bile tract surgery' OR 'biliary surgery' OR 'biliary tract operation' OR 'biliary tract reoperation' OR 'biliary tract surgery' OR 'biliary tract surgical procedures' OR 'gall bladder surgery' OR 'gallbladder operation' OR 'gallbladder surgery' OR 'sphincterotomy, transhepatic' OR 'surgery, biliary tract'	**89,610**
C	'conservative management' OR 'conservative therapy' OR 'conservative treatment' OR 'nonoperative treatment' OR 'nonsurgical treatment' OR 'organ sparing treatment' OR 'organ sparing treatments' OR 'treatment, conservative' OR analges* OR 'non-steroid anti-inflammatory' OR 'non-steroidal anti-inflammatory' OR nsaid OR 'anti inflammatory' OR antinflammatory OR antibiotic* OR supportive OR 'non invasive' OR 'non surgical' OR 'non operative' OR 'wait and watch' OR 'wait and see' OR 'watchful waiting' OR 'usual care'	**1,783,181**
S	trial OR trail OR blind OR rct OR randomization OR randomisation OR randomized OR randomised OR placebo OR 'randomized control trial' OR 'randomized controlled trial' OR 'randomized controlled trail' OR 'randomized control trail' OR 'controlled clinical trial' OR randomly OR groups OR 'randomised controlled study' OR 'randomized controlled study'	**5,139,403**
PICS		**1646**

Supplementary Tables

Supplementary Table 7. Search terms and no. of hits in Scopus

PICOS	**SCOPUS Search terms**	**Hits (on 16th July 2020)**
P	ALL(“gallbladder" OR gallbladder OR cholecystitis OR cholecystitis OR cholelithiasis OR cholelithiasis OR "gall stone" OR gallstone OR cholecystolithiasis OR calculi OR choledocholithiasis OR choledocholithiasis OR "bile duct stone" OR "cbd stone" OR "biliary disease" OR "biliary tract disease" OR “Gallbladder inflammation”)	**630,221**
I	ALL (cholecystectomy OR "gallbladder removal" OR "gallbladder excision" OR ("surgical removal" AND ("gall bladder" OR gallbladder)) OR ("surgical excision" AND ("gall bladder" OR gallbladder)) OR "restrictive strategy" OR cholecystostomy OR "biliary tract operation" OR "biliary tract reoperation" OR "biliary tract surgical procedures" OR "gallbladder operation" OR sphincterotomy OR (surger* AND (gall OR biliar*))	**283,560**
C	ALL (conservative OR observatio* OR analgesi* OR (nonsteroid AND anti-inflammatory) OR "non-steroidal anti-inflammatory" OR nsaid OR anti-inflammatory OR antinflammatory OR anti-infective OR anti-bacterial OR supportive OR non-invasive OR noninvasive OR non-surgical OR nonsurgical OR non-operative OR nonoperative OR "wait and watch" OR "wait and see" OR watchful-waiting OR "usual care")	**9,204,582**
S	ALL (trial OR trail OR blind OR rct OR randomization OR randomisation OR randomized OR randomised OR placebo OR 'randomized control trial' OR 'randomized controlled trial' OR 'randomized controlled trail' OR 'randomized control trail' OR 'controlled clinical trial' OR randomly OR groups OR 'randomised controlled study' OR 'randomized controlled study')	**1,184,853**
PICS	Limited only to journal	**2.903**

Supplementary Tables

References :

1. Agrawal R, Sood KC, Agarwal B. Evaluation of Early versus Delayed Laparoscopic Cholecystectomy in Acute Cholecystitis. *Surg Res Pract* 2015;2015:349801.
2. Ammar SA, Bar MA, El Shafy M. Laparoscopic cholecystectomy after endoscopic retrograde cholangiopancreatography: The optimal timing for operation. *World Journal of Laparoscopic Surgery* 2014;7(2):69-73.
3. El Nakeeb A, Ezzet H, Askar W, et al. Early Versus Late Cholecystectomy after Clearance of Common Bile Duct Stones by Endoscopic Retrograde Cholangiopancreatography: A Prospective Randomized Study. *Surgical Laparoscopy, Endoscopy and Percutaneous Techniques* 2016;26(3):202-207.
4. Zahur S, Rabbani S, Andrabi SIH, et al. Early vs interval cholecystectomy in acute cholecystitis: An experience at Ghurki Trust Teaching Hospital, Lahore. *Pakistan Journal of Medical and Health Sciences* 2014;8(3):778-781.
5. Gul R, Dar RA, Sheikh RA, et al. Comparison of early and delayed laparoscopic cholecystectomy for acute cholecystitis: Experience from a single center. *North American Journal of Medical Sciences* 2013;5(7):414-418.
6. Gutt CN, Encke J, Köninger J, et al. Acute cholecystitis: early versus delayed cholecystectomy, a multicenter randomized trial (ACDC study, NCT00447304). *Ann Surg* 2013;258(3):385-393.
7. Järvinen HJ, Hästbacka J. Early cholecystectomy for acute cholecystitis: a prospective randomized study. *Ann Surg* 1980;191(4):501-505.
8. Khalid S, Iqbal Z, Bhatti AA. Early Versus Delayed Laparoscopic Cholecystectomy For Acute Cholecystitis. *Journal of Ayub Medical College, Abbottabad : JAMC* 2017;29(4):570-573.
9. Kolla SB, Aggarwal S, Kumar A, et al. Early versus delayed laparoscopic cholecystectomy for acute cholecystitis: a prospective randomized trial. *Surg Endosc*;18(9):1323-1327.
10. Zhang J, Li NP, Huang BC, et al. The Value of Performing Early Non-enhanced CT in Developing Strategies for Treating Acute Gallstone Pancreatitis. *J Gastrointest Surg*;20(3):604-610.
11. Lo CM, Liu CL, Fan ST, et al. Prospective randomized study of early versus delayed laparoscopic cholecystectomy for acute cholecystitis. *Annals of Surgery* 1998;227(4):461-467.
12. Lahtinen J, Alhava EM, Aukee S. Acute cholecystitis treated by early and delayed surgery. A controlled clinical trial. *Scand J Gastroenterol* 1978;13(6):673-678.
13. Norby S, Herlin P, Holmin T. Early or delayed cholecystectomy in acute cholecystitis? A clinical trial. *British Journal of Surgery* 1983;70(3):163-165.
14. da Costa DW, Bouwense SA, Schepers NJ, et al. Same-admission versus interval cholecystectomy for mild gallstone pancreatitis (PONCHO): a multicentre randomised controlled trial. *Lancet* 2015;386(10000):1261-1268.
15. McArthur P, Cuschieri A, Sells RA, et al. Controlled clinical trial comparing early with interval cholecystectomy for acute cholecystitis. *Br J Surg* 1975;62(10):850-852.
16. Rajcok M, Bak V, Danihel L, et al. Early versus delayed laparoscopic cholecystectomy in treatment of acute cholecystitis. *Bratislavske lekarske listy* 2016;117(6):328-331.
17. Saber A, Hokkam EN. Operative outcome and patient satisfaction in early and delayed laparoscopic cholecystectomy for acute cholecystitis. *Minim Invasive Surg* 2014;2014:162643.
18. Lai PB, Kwong KH, Leung KL, et al. Randomized trial of early versus delayed laparoscopic cholecystectomy for acute cholecystitis. *Br J Surg*;85(6):764-767.
19. Macafee DA, Humes DJ, Bouliotis G, et al. Prospective randomized trial using cost-utility analysis of early versus delayed laparoscopic cholecystectomy for acute gallbladder disease. *Br J Surg* 2009;96(9):1031-1040.
20. Noel R, Arnelo U, Lundell L, et al. Index versus delayed cholecystectomy in mild gallstone pancreatitis: results of a randomized controlled trial. *HPB (Oxford)* 2018;20(10):932-938.

Supplementary Tables

21. Reinders JS, Goud A, Timmer R, et al. Early laparoscopic cholecystectomy improves outcomes after endoscopic sphincterotomy for choledochocystolithiasis. *Gastroenterology* 2010;138(7):2315-2320.
22. Roulin D, Saadi A, Di Mare L, et al. Early versus delayed cholecystectomy for acute cholecystitis, are the 72 hours still the rule? A randomized trial. *Annals of Surgery* 2016;264(5):717-722.
23. Stone HH, Fabian TC, Dunlop WE. Gallstone pancreatitis: biliary tract pathology in relation to time of operation. *Ann Surg*;194(3):305-312.
24. Jee SL, Jarmin R, Lim KF, et al. Outcomes of early versus delayed cholecystectomy in patients with mild to moderate acute biliary pancreatitis: A randomized prospective study. *Asian J Surg* 2018;41(1):47-54.
25. Salman B, Yilmaz U, Kerem M, et al. The timing of laparoscopic cholecystectomy after endoscopic retrograde cholangiopancreaticography in cholelithiasis coexisting with choledocholithiasis. *J Hepatobiliary Pancreat Surg* 2009;16(6):832-836.
26. Yadav RP, Adhikary S, Agrawal CS, et al. A comparative study of early vs. delayed laparoscopic cholecystectomy in acute cholecystitis. *Kathmandu University Medical Journal* 2009;7(25):16-20.
27. Abbas A, Umair M, Isnain HG, et al. Early cholecystectomy vs delayed cholecystectomy for gall stone pancreatitis. *Pakistan Journal of Medical and Health Sciences* 2013;7(1):87-89.
28. Ozkardeş AB, Tokaç M, Dumlu EG, et al. Early versus delayed laparoscopic cholecystectomy for acute cholecystitis: a prospective, randomized study. *Int Surg* 2014;99(1):56-61.
29. Verma S, Agarwal PN, Bali RS, et al. Early versus Delayed Laparoscopic Cholecystectomy for Acute Cholecystitis: A Prospective Randomized Trial.
30. Aboulian A, Chan T, Yaghoubian A, et al. Early cholecystectomy safely decreases hospital stay in patients with mild gallstone pancreatitis: A randomized prospective study. *Annals of Surgery* 2010;251(4):615-619.
31. Hammarstr ├Ц m LE, Holmin T, Stridbeck H, et al. Long-term follow-up of a prospective randomized study of endoscopic versus surgical treatment of bile duct calculi in patients with gallbladder in situ. *Br J Surg* 1995;82(11):1516-1521.
32. Targarona EM, Ayuso RM, Bordas JM, et al. Randomised trial of endoscopic sphincterotomy with gallbladder left in situ versus open surgery for common bileduct calculi in high-risk patients. *Lancet* 1996;347(9006):926-929.
33. Boerma D, Rauws EA, Keulemans YC, et al. Wait-and-see policy or laparoscopic cholecystectomy after endoscopic sphincterotomy for bile-duct stones: a randomised trial. *Lancet*;360(9335):761-765.
34. Lau JYW, Leow C-K, Fung TMK, et al. Cholecystectomy or gallbladder in situ after endoscopic sphincterotomy and bile duct stone removal in Chinese patients. *Gastroenterology* 2006;130(1):96-103.
35. Schmidt M, Søndenaa K, Vetrhus M, et al. A Randomized Controlled Study of Uncomplicated Gallstone Disease with a 14-Year Follow-Up Showed that Operation Was the Preferred Treatment. *Digestive Surgery* 2011;28(4):270-276.
36. Schmidt M, Søndenaa K, Vetrhus M, et al. Long-term follow-up of a randomized controlled trial of observation versus surgery for acute cholecystitis: Non-operative management is an option in some patients. *Scandinavian Journal of Gastroenterology* 2011;46(10):1257-1262.
37. Zargar SA, Mushtaq M, Beg MA, et al. Wait-and-see policy versus cholecystectomy after endoscopic sphincterotomy for bile-duct stones in high-risk patients with co-existing gallbladder stones: a prospective randomised trial. *Arab Journal of Gastroenterology: The Official Publication of the Pan-Arab Association of Gastroenterology* 2014;15(1):24-26.
38. Suc B, Escat J, Cherqui D, et al. Surgery vs endoscopy as primary treatment in symptomatic patients with suspected common bile duct stones: a multicenter randomized trial. French Associations for Surgical Research. *Arch Surg* 1998;133(7):702-708.

Supplementary Tables

39. Heo J, Jung MK, Cho CM. Should prophylactic cholecystectomy be performed in patients with concomitant gallstones after endoscopic sphincterotomy for bile duct stones? *Surgical Endoscopy* 2015;29(6):1574-1579.

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Yes, 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Yes, 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Yes, 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Yes, 4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Yes, 4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Yes, 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Yes, 5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Yes, 4 & appendix A
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Yes, 5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Yes, 6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Yes, 6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Yes, 6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Yes, 7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Yes, 7

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Yes 8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Yes, 7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Yes, 8 & 9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Yes, 9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Yes, 9& 10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Yes, 10-15 Figure 2 & 3, Table 1&2, Supp figures 1-20
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Yes, 10-15
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Yes, 10-15 Supp Figures 4,14,17,20
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Supp Table 1&2, Supp Figures 2-11, 13, 15,16,18,19
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Yes, 16 & 17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Yes, 18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Yes, 18
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Yes, 1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097